

PAPER NO. 2103 - SELECTED QUESTIONS

UNIT 1 | UNIT 2 | UNIT 7

Question numbers follow the source paper.

UNIT 1 - STATICS OF A PARTICLE

5. A particle of mass 50 lb is suspended by two strings 3 ft and 4 ft attached to points at the same level, whose distance apart is 5 ft. Find the tensions in the strings.
6. A mass of 90 lb is suspended by two light strings of 9 ft and 12 ft from two points at the same level and 15 ft apart. Find the tensions in the strings.
7. A string ABCD is suspended from two points A and D at the same level. A weight of 10 lb is hung from a string knotted to ABCD at B, and a weight w lb is hung from a string knotted to ABCD at C. If AB and CD make angles of 45° and 60° respectively with the horizontal, and if BC is horizontal, find the tension in the string BC and the weight w .
10. A particle is acted on by forces of 1, 2, 3 and 4 lb. wt., the angles between them being 60° , 30° , 60° respectively. Find the magnitude and direction of the resultant.
11. Forces of 4, 3, 2, and 1 lb. wt. act at a point A in directions AB, AC, AD and AE where angle BAC = 30° , angle CAD = 30° , angle DAE = 90° . Find the magnitude of their resultant and the inclination of its direction to AB. Verify by drawing a vector-diagram.

UNIT 2 - STATICS OF A RIGID BODY - PARALLEL FORCES, MOMENTS, COUPLES

1. Like parallel forces of 9 and 12 lb. wt. act at points A and B which are 42 in apart. Find the magnitude of their resultant and the point in which it cuts AB. Find also the magnitude and position of the resultant when the forces are unlike.
3. A uniform rod AB, of length 12 ft and weighing 50 lb, rests on two supports, one at A and the other 2 ft from B. Weights of 4, 5 and 10 lb are attached at points 2 ft, 4 ft and 8 ft respectively from A. Find the thrusts on the supports.
4. A uniform beam AB, 20 ft long and weighing 40 lb, rests on two supports, one at A and the other 4 ft from B. If a weight of 20 lb is attached to the beam at a point 12 ft from A, find the pressures on the supports.
5. Weights of 1 kg, 2 kg, 3 kg, 4 kg and 5 kg are suspended from a uniform rod AB, 6 m long, weighing 3 kg and supported at its ends, at distances of 1 m, 2 m, 3 m, 4 m, 5 m from A. Find the pressures on the supports.
7. A uniform rod AB, 12 ft long and weighing 50 lb, is pivoted at a point 3 ft from A. A weight of 200 lb is suspended from A. What force applied at B, in a direction perpendicular to the rod, will keep it in equilibrium with A below B and AB inclined at 60° to the horizontal?

UNIT 7 - SPEED AND VELOCITY

1. A boat is towed with a velocity of 4 m.p.h. straight across a river which is flowing at 3 m.p.h. Find the magnitude and direction of the resultant velocity of the boat. If the breadth of the river is 400 ft, find how far down the river the boat will reach the opposite bank.
3. A ship is steaming due east at 15 m.p.h. and another ship is steaming due south at 20 m.p.h. Find the velocity of the second ship relative to the first.
4. One ship is sailing due east at 12 m.p.h. and another ship is sailing due north at 16 m.p.h. Find the velocity of the second ship relative to the first.
5. A train is travelling along a horizontal rail at 45 m.p.h. and rain is falling vertically with a velocity of 22 ft. per sec. Find the apparent direction and velocity of the rain to a person travelling in the train.
6. Raindrops are falling through the air with a velocity of 10 ft. per sec. If a north wind blows at 12 m.p.h., find the direction in which the drops appear to fall to a person walking north at 3 m.p.h. With what velocity would they hit his umbrella?